SUPPLEMENTAL MATERIAL

Table S1. PRISMA 2009 Checklist.

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	n/a
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10-11
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10-11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	18

Table S2. Literature search strategy.

Database	Search strategies	Results
PubMed	(("air pollution"[Mesh Terms] OR "air pollution"[All Fields]) OR ("particulate matter"[Mesh Terms] OR "particulate matter"[All Fields]) OR "particles"[All Fields] OR ("air pollutants"[Mesh Terms] OR "air pollutants"[All Fields]) OR "fine particulate" [All Fields] OR ("sulfur dioxide"[Mesh Terms] OR "sulfur dioxide"[All Fields]) OR ("sulfur oxide" [Mesh Terms] OR " sulfur oxide"[All Fields]) OR ("nitrogen dioxide"[Mesh Terms] OR "nitrogen dioxide"[All Fields]) OR("nitrogen oxide"[Mesh Terms] OR "nitrogen oxide"[All Fields]) OR ("ozone"[Mesh Terms] OR "ozone"[All Fields]) OR ("carbon monoxide"[Mesh Terms] OR "carbon monoxide"[All Fields]) OR ("black carbon" [Mesh Terms] OR "black carbon"[All Fields]) OR "$PM_{2.5}$"[All Fields] OR "PM_{10}"[All Fields] OR "SO_2"[All Fields] OR "SO_X"[All Fields] OR "NO_2"[All Fields] OR "NO_X"[All Fields] OR "O_3"[All Fields] OR "CO"[All Fields] OR "BC"[All Fields]) AND (("hypertension"[Mesh Terms] OR "hypertension"[All Fields]) OR "high blood pressure"[All Fields] OR hypertensive[All Fields] OR ("blood pressure"[Mesh Terms] OR "blood pressure"[All Fields]) OR "systolic blood pressure"[All Fields] OR "diastolic blood pressure"[All Fields]) OR ("hypotension" [Mesh Terms] OR "hypotension"[All Fields]) OR "hypotensive"[All Fields] OR ("arterial pressure" [Mesh Terms] OR "arterial pressure"[All Fields]) OR ("arterial tension" [Mesh Terms] OR "arterial tension"[All Fields]) OR ("elevated blood pressure" [Mesh Terms] OR "elevated blood pressure"[All Fields])) AND (("child"[Mesh Terms] OR "child"[All Fields]) OR ("children"[Mesh Terms] OR "children"[All Fields]) OR "childhood"[All Fields] OR ("adolescent"[Mesh Terms] OR "adolescent"[All Fields]) OR ("teenager" [Mesh Terms] OR "teenager"[All Fields]) OR "kids"[All Fields])	19689 records
Embase	('air pollution'/exp OR 'air pollution' OR 'particulate matter'/exp OR 'particulate matter' OR 'particles' OR 'air pollutants'/exp OR 'air pollutants' OR 'fine particulate' OR 'sulfur dioxide'/exp OR 'sulfur dioxide' OR 'sulfur oxide'/exp OR 'sulfur oxide' OR 'nitrogen dioxide'/exp OR 'nitrogen dioxide' OR 'nitrogen oxide'/exp OR 'nitrogen oxide' OR 'ozone'/exp OR ozone OR 'carbon monoxide'/exp OR 'carbon monoxide' OR 'black carbon'/exp OR 'black carbon' OR '$PM_{2.5}$' OR 'PM_{10}' OR 'SO_2' OR 'SO_X' OR 'NO_2' OR 'NO_X' OR 'O_3' OR 'CO' OR 'BC') AND ('hypertension'/exp OR hypertension OR 'high blood pressure'/exp OR 'high blood pressure' OR hypertensive OR 'blood pressure'/exp OR 'blood pressure' OR 'systolic blood pressure'/exp OR 'systolic blood pressure' OR 'diastolic blood pressure'/exp OR 'diastolic blood pressure' OR 'hypotension'/exp OR 'hypotension' OR	14878 records

	'hypotensive' OR 'arterial pressure'/exp OR 'arterial pressure' OR 'arterial tension' OR 'arterial tension' OR 'elevated blood pressure' OR 'elevated blood pressure') AND ('child'/exp OR 'child' OR 'children'/exp OR 'children' OR 'childhood' OR 'adolescents'/exp OR 'adolescents' OR 'teenager'/exp OR 'teenager' OR 'kids')	
Web of Science	TS=(“air pollution” OR "particulate matter" OR "particles" OR "air pollutants" OR "fine particulate" OR "sulfur dioxide" OR "sulfur oxide" OR "nitrogen dioxide" OR "nitrogen oxide" OR "ozone" OR "carbon monoxide" OR "black carbon" OR "$PM_{2.5}$" OR "PM_{10}" OR "SO_2" OR "SO_X" OR "NO_2" OR "NO_X" OR "O_3" OR "CO" OR "BC") AND TS=("hypertension" OR "high blood pressure" OR "hypertensive" OR "blood pressure" OR "systolic blood pressure" OR "diastolic blood pressure" OR "hypotension" OR "hypotensive" OR "arterial pressure" OR "arterial tension" OR "elevated blood pressure") AND TS=(“child” OR “children” OR “childhood” OR “adolescents” OR “teenager” OR “kids”)	2083 records

Table S3. Contextual details of included studies: exposure and blood pressure measurements and confounding factors.

Author (Publication year)	Subjects	Pollutants	NO_2	$PM_{2.5}$	PM_{10}	Exposure assessment method	Confounding factors
Zhang et al.(2020)[18]	Children from five schools in five districts of Guangzhou	$PM_{2.5}$ PM_{10}	-	39.00	61.10	Monitoring stations and inverse distance weighting interpolation method	age, sex, weight status, humidity, breast, outdoor physical activity time, parental education, parental smoking status and family history of hypertension
Wu et al.(2020)[19]	Children from the Chinese Seven Northeast Cities (SNEC) Study	$PM_{2.5}$	-	54.16	-	Monitoring station	age, sex, parental education, income, passive tobacco smoke exposure,

							home coal use, exercise time, and BMI.
Warembourg et al.(2019)[20]	Child from the HELIX (Human Early-Life Exposome) project	NO_2 $PM_{2.5}$ PM_{10}	22.10	13.50	25.60	Land use regression or dispersion models	age, sex, height, maternal age, maternal education level, maternal pre-pregnancy body mass index, parity, parental country of birth and cohort
Ntarladima et al.(2019)[21]	Children from the Wheezing Illnesses Study Leidsche Rijn (WHISTLER)	NO_2 $PM_{2.5}$ PM_{10}	29.40	16.70	25.00	Land use regression model	sex, age, parental socio-economic status characteristics, exposed to smoke during pregnancy, child exposed to smoke later in life
Yang et al.(2019)[9]	Children from the 2016 Health Promotion Program for Children and Adolescents (HPPCA)	$PM_{2.5}$ PM_{10}	-	45.70	76.20	Monitoring station	gender, age, body mass index, outdoor temperature, O3

Zhang et al.(2019)[8]	Children and adolescents from seven provinces/municipalities (Chongqing, Hunan, Guangdong, Liaoning, Ningxia, Shanghai and Tianjin) in China	$PM_{2.5}$ PM_{10}	-	60.10	99.40	Machine learning method with Radom Forests model	age, sex, height, body mass index, exercise, dietary, parental education, parental smoking, parental hypertension, region (south vs north), rurality (urban vs rural),annual average temperature and GDP per capita
Li et al(2018)[22]	Primary or middle school students from the 2010 Chinese National Survey on Students' Constitution and Health(CNSSCH)	PM_{10}	-	-	97.00	Monitoring station	effect of school, age, gender, height, weight of each child, GDP per capital, relative humidity in 2010 of each city, the spline term of temperature, NO2, SO2, children's physical activity level

Zeng et al(2017)[7]	Children from the Chinese Seven Northeast Cities (SNEC) Study	PM_{10}	-	-	108.80	Monitoring station	temperature, age, sex, BMI, breast feeding, birth weight, exercise time, personal area, passive smoking exposure, parental education, family income, family history of hypertension, and district.
Piters et al.(2015)[11]	Primary schools students from Health Effects of Air Pollution in Antwerp Schools(HEAPS) study	$PM_{2.5}$ PM_{10}		35.00	24.00	Air pollution monitoring devices	sex, age, height and weight of the child, parental education, neighborhood SES, fish consumption, heart rate, school, day of the week, season, wind, speed, relative humidity, temperature

Bilenko et al.(2015)[6]	Children from the Prevention and Incidence of Asthma and Mite Allergy (PIAMA) birth cohort study	NO_2 $PM_{2.5}$ PM_{10}	13.3（short-term） 7.8(long-term)	1.1(long-term)	11.1(short-term) 1(long-term)	Land use regression model	sex, age, height, and BMI, cuff size, gestational age at birth, birthweight, weight gain during the first year of life, breast feeding, maternal smoking during pregnancy, parental smoking in child's home, physical activity, puberty development scale, maternal education, maternal hypertension during pregnancy, pneumonia and/or otitis media during the first 2 years of life, ambient temperature, and

							room temperature
Dong et al.(2014)[23]	Children from the Seven Northeastern Cities Chinese Children's Study (SNECCS)	NO_2 PM_{10}	36.44	-	88.90	Monitoring station	age, sex, BMI, parental education, low birth weight, premature birth, breast, income, passive smoking exposure, home coal use, exercise time, area residence per person, family history of hypertension, and district

Liu et al.(2014)[10]	Children from the German Infant Nutritional Intervention plus environmental and genetic influences on allergy development study (GINIplus)	NO_2 $PM_{2.5}$ PM_{10}	23.31	14.88	22.12	Land use regression model	cohort study, area, gender, age of child, BMI, physical activity, maternal smoking during pregnancy, parental education level, parental history of hypertension,7-day level of air pollutants, 7-day temperature
Baumgartner et al.(2012)[24]	Children from villages in Yunnan, China	$PM_{2.5}$	-	53.00	-	Portable, battery-operated pump	sex, age, height, body mass index, passive smoking, socioeconomic status, salt intake, monosodium glutamate use, physical activity

Clark et al.(2012)[25]	Children from the Road Traffic and Aircraft Noise Exposure and Children's Cognition and Health (RANCH) project	NO_2	42.73	-	-	Combined emission-dispersion and regression model	age, gender, employment status, crowding, home ownership, mother's educational level, long-standing illness, main language spoken at home, parental support for schoolwork, body mass index, cuff-size, room temperature, birth weight, parental high blood pressure, and prematurity.

Abbreviations: O_3, ozone; GDP, Gross Domestic Product; SES, socio-economic status; NO_2, nitrogen dioxide; SO_2, sulfur dioxide; BMI, body mass index. The levels of pollutants are present with mean/interquartile range: Piters et al.(2015) and Bilenko et al.(2015) are present with interquartile range and the rest are present with mean.

Table S4. Results of the sensitivity analysis for short-term exposure to PM_{10} and change in DBP.

Excluded article	Estimated effect	95% CI	P	I^2	P of publication bias
Yang et al.(2019)[9]	0.051	-0.506, 0.608	0.858	84.23%	0.001
Zeng et al(2017)[7]	0.041	-0.407, 0.49	0.857	76.15%	0.007
Pieters et al.(2015)[11]	0.327	0.16, 0.493	<0.001	93.35%	0.120
Bilenko et al.(2015)[6]	0.358	0.229, 0.487	<0.001	87.59%	0.039

Table S5. Results of the sensitivity analysis for long-term exposure to $PM_{2.5}$ and change in SBP.

Excluded article	Estimated effect	95%CI	P	I^2	P of publication bias
Zhang et al.(2020)[18]	1.978	1.114,2.841	<0.001	0.00%	0.850
Wu et al.(2020)[19]	1.268	0.097,2.439	0.03	0.00%	0.702
Warembourg et al.(2019)[20]	1.826	0.951,2.701	<0.001	10.49%	0.355
Ntarladima et al.(2019)[21]	1.778	0.924,2.631	<0.001	5.90%	0.177
Zhang et al.(2019)[8]	2.003	0.946,3.059	<0.001	4.72%	0.269
Bilenko et al.(2015)[6]	1.84	0.988,2.692	<0.001	4.04%	0.4787
Liu et al.(2014)[10]	1.786	0.925,2.647	<0.001	9.57%	0.2525

Table S6. Results of the sensitivity analysis for long-term exposure to $PM_{2.5}$ and change in DBP.

Excluded article	Estimated effect	95%CI	P	I^2	P of publication bias
Zhang et al.(2020)[18]	0.952	0.156,1.748	0.0191	1.89%	0.1128
Wu et al.(2020)[19]	1.041	0.069,2.012	0.0357	0.20%	0.1714
Warembourg et al.(2019)[20]	0.835	0.037,1.633	0.0404	0.00%	0.2433
Ntarladima et al.(2019)[21]	0.867	0.087,1.646	0.0293	0.00%	0.4043
Zhang et al.(2019)[8]	1.048	0.017,2.080	0.0464	0.62%	0.1588
Bilenko et al.(2015)[6]	0.86	0.079,1.641	0.0310	0.00%	0.3735
Liu et al.(2014)[10]	1.002	0.209,1.795	0.0133	0.00%	0.0668

Table S7. Results of the sensitivity analysis for long-term exposure to PM_{10} and change in SBP.

Excluded article	Estimated effect	95%CI	P	I^2	P of publication bias

Zhang et al.(2020)[18]	0.529	0.074,0.983	0.0226	93.11%	0.3685
Warembourg et al.(2019)[20]	0.548	0.066,1.030	0.0257	93.68%	0.3556
Ntarladima et al.(2019)[21]	0.516	0.085,0.948	0.0189	92.57%	0.5363
Zhang et al.(2019)[8]	0.409	-0.007,0.826	0.0540	90.68%	0.5709
Li et al(2018)[22]	0.692	0.573,0.811	0.0000	0.00%	0.6841
Bilenko et al.(2015)[6]	0.529	0.094,0.964	0.0173	92.72%	0.2996
Dong et al.(2014)[23]	0.079	0.062,0.096	0.0000	20.54%	0.0429
Liu et al.(2014)[10]	0.538	0.064,1.012	0.0262	93.52%	0.3705

Table S8. Results of the sensitivity analysis for long-term exposure to PM_{10} and change in DBP.

Excluded article	Estimated effect	95%CI	P	I^2	P of publication bias
Zhang et al.(2020)[18]	0.387	0.009,0.766	0.045	92.50%	0.4827
Warembourg et al.(2019)[20]	0.359	-0.027,0.744	0.068	92.60%	0.6129
Ntarladima et al.(2019)[21]	0.37	0.013,0.727	0.042	91.88%	0.8301
Zhang et al.(2019)[8]	0.332	-0.068,0.731	0.104	92.67%	0.6291
Li et al(2018)[22]	0.626	0.524,0.727	0.000	0.00%	0.8247
Bilenko et al.(2015)[6]	0.366	0.008,0.725	0.045	91.91%	0.7731
Dong et al.(2014)[23]	0.081	0.067,0.095	0.000	0.00%	0.1672
Liu et al.(2014)[10]	0.435	0.068,0.802	0.020	91.91%	0.2444

Table S9. Results of the sensitivity analysis for long-term exposure to NO_2 and change in SBP.

Excluded article	Estimated effect	95%CI	P	I^2	P of publication bias
Warembourg et al.(2019)[20]	0.436	-0.156, 1.028	0.1488	53.14%	0.897
Ntarladima et al.(2019)[21]	0.487	-0.001,0.976	0.0506	51.81%	0.262
Bilenko et al.(2015)[6]	0.824	0.603,1.045	<0.001	0.00%	0.682
Dong et al.(2014)[23]	0.228	-0.251,0.707	0.3508	0.00%	0.311
Liu et al.(2014)[10]	0.791	0.571,1.012	<0.001	42.48%	0.866
Clark et al.(2012)[25]	0.481	-0.057,1.019	0.0796	53.82%	0.160

Table S10. Results of the sensitivity analysis for long-term exposure to NO_2 and change in DBP.

Excluded article	Estimated effect	95%CI	P	I2	P of publication bias
Warembourg et al.(2019)[20]	0.408	-0.133,94.90%	0.139	59.72%	0.7251

Ntarladima et al.(2019)[21]	0.371	-0.095,83.80%	0.119	60.65%	0.0154
Bilenko et al.(2015)[6]	0.439	-0.095,97.30%	0.108	57.09%	0.7392
Dong et al.(2014)[23]	0.090	-0.320,50.10%	0.667	0.00%	0.2082
Liu et al.(2014)[10]	0.800	0.613,98.70%	0.000	39.85%	0.189
Clark et al.(2012)[25]	0.384	-0.137,90.40%	0.148	61.59%	0.7535

Table S11 Subgroup analysis for the association between long-term exposure and SBP

Air pollutant	design	No. of estimates	Sample size	I^2	P_{egger}	P	Estimated effect (95%CI)
NO_2							
	cross-sectional	3	10363	0.00%	0.874	<0.001	0.880 (0.644,1.115)
	cohort	3	5045	8.56%	0.141	0.5023	0.173 (-0.333,0.680)
$PM_{2.5}$							
	cross-sectional	4	61057	41.29%	0.322	<0.001	1.837 (0.940,2.734)
	cohort	3	5045	0.00%	0.692	0.2284	1.575 (0.988,4.134)
PM_{10}							
	cross-sectional	5	132820	96.55%	0.2179	0.0439	0.575(0.016,1.135)
	cohort	3	5045	0	0.9543	0.3376	0.466(-0.487,1.419)

Table S12 Subgroup analysis for the association between long-term exposure and DBP

Air pollutant	design	No. of estimates	Sample size	I^2	P_{egger}	P	Estimated effect (95%CI)
NO_2							
	cross-sectional	3	10363	0.00%	0.795	<0.001	0.876 (0.660,1.093)
	cohort	3	5045	0.00%	0.973	0.7873	0.066(-0.412,0.543)
$PM_{2.5}$							
	cross-sectional	4	61057	0.00%	0.304	0.0509	0.825 (-0.003,1.653)
	cohort	3	5045	26.01%	0.392	0.1343	1.657 (-0.512,3.825)
PM_{10}							
	cross-sectional	5	132820	94.88	0.398	0.0442	0.408(0.011,0.806)
	cohort	3	5045	23.94	0.5719	0.6735	0.198(-0.722,1.117)

Table S13. The results of univariate meta-regression in the long-term exposure group.

	Age		Male proportion		Study location	
	β(95%CI)	P	β(95%CI)	P	β(95%CI)	P
SBP						
NO_2	-0.176(-0.514,0.162)	0.308	0.066(-0.148, 0.281)	0.545	-0.657(-1.192,-0.122)	0.016

$PM_{2.5}$	0.043(-0.716,0.802)	0.912	-0.302(-1.034,0.431)	0.420	0.045(-2.506,2.595)	0.973
PM_{10}	-0.361(-0.433,-0.288)	<0.001	0.512(0.403,0.621)	<0.001	0.423(-0.520,1.366)	0.379
DBP						
NO_2	0.085(-0.151,0.321)	0.479	0.018(-0.133,0.170)	0.814	-0.794(-1.251,-0.338)	0.001
$PM_{2.5}$	-0.343(-1.039,0.352)	0.333	0.123(-0.509,0.754)	0.704	1.282(-0.938,3.502)	0.258
PM_{10}	-0.322(-0.384,-0.260)	<0.001	0.461(0.368,0.555)	<0.001	0.165(-0.745,1.076)	0.722

Table S14. The results of multivariate meta-regression in the long-term exposure group.

	Age		Male proportion		Study location	
	β(95%CI)	P	β(95%CI)	P	β(95%CI)	P
SBP						
NO_2	-0.266(-0.625,0.093)	0.308	-0.051(-0.293,0.192)	0.683	-0.800(-1.377,-0.223)	0.007
$PM_{2.5}$	-0.481(-1.908,0.946)	0.912	-0.695(-1.874,0.484)	0.248	0.197(-3.289,3.683)	0.912
PM_{10}	-0.693(-1.093,-0.293)	<0.001	-0.493(-1.109,0.122)	0.116	-0.562(-1.611,0.487)	0.293
DBP						
NO_2	-0.088(-0.406,0.231)	0.479	-0.044(-0.240,0.152)	0.663	-0.860(-1.372,-0.348)	0.001
$PM_{2.5}$	-0.493(-1.855,0.869)	0.333	-0.343(-1.442,0.755)	0.540	0.856(-1.920,3.633)	0.546
PM_{10}	-0.536(-0.887,-0.185)	<0.001	-0.309(-0.849,0.231)	0.262	-0.785(-1.796,0.226)	0.128

Figure S1. Forest plot for the association between short-term exposure to PM_{10} (per 10 μg/m^3 increment) and SBP (mmHg). The blue boxes represent estimated effects of included studies. The blue horizontal bars represent 95% CIs of the estimated effects. The green box represents the pooled mean estimated effect. CI indicates confidence interval.

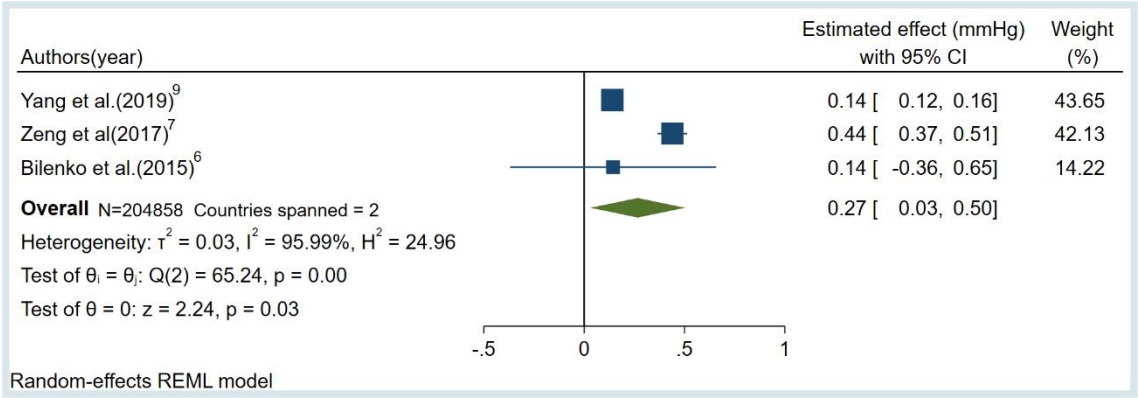


Figure S2. Forest plot for the association between short-term exposure to $PM_{2.5}$ (per 10 μg/m^3 increment) and DBP (mmHg).

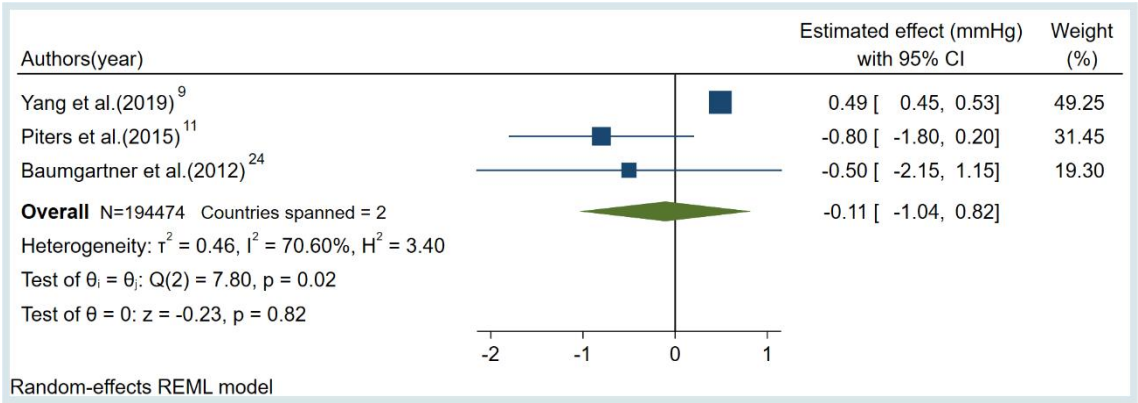


Figure S3. Forest plot for the association between short-term exposure to PM_{10} (per 10 μg/m^3 increment) and DBP(mmHg).

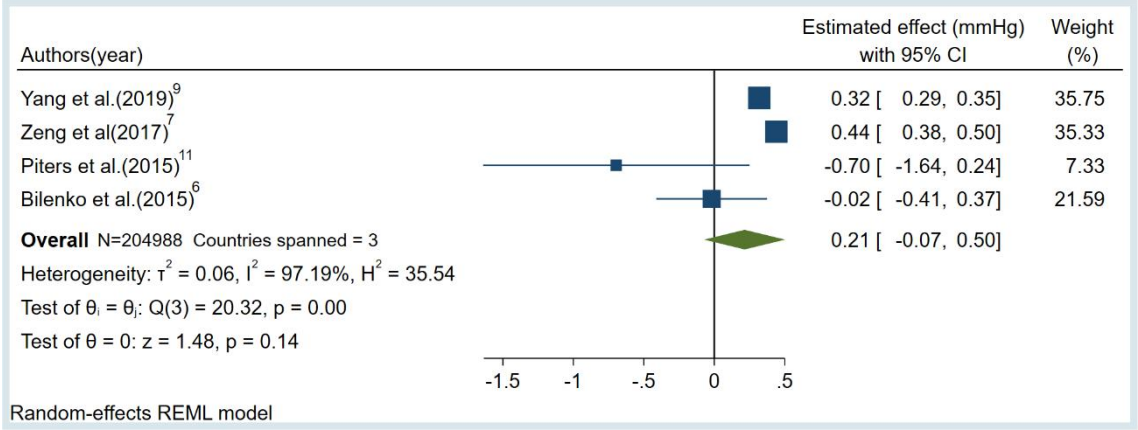


Figure S4. Funnel plot analysis in the meta-analysis of the association between short-term exposure to PM_{10} and SBP. The x-coordinate is the estimated effect and the y-coordinate is the standard error of estimated effect. The two diagonal lines are the edges of the funnel plot. The black dots represent studies that were included in the meta-analysis. The study with small simple size is at the bottom of funnel plot, while the study with large sample is at the top of funnel plot.

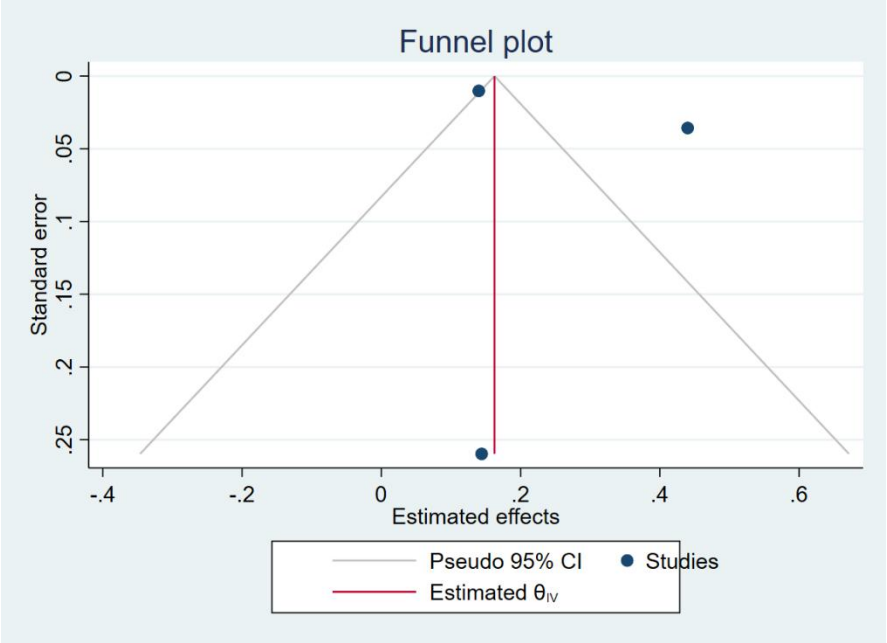


Figure S5. Funnel plot analysis in the meta-analysis of the association between short-term exposure to $PM_{2.5}$ and DBP.

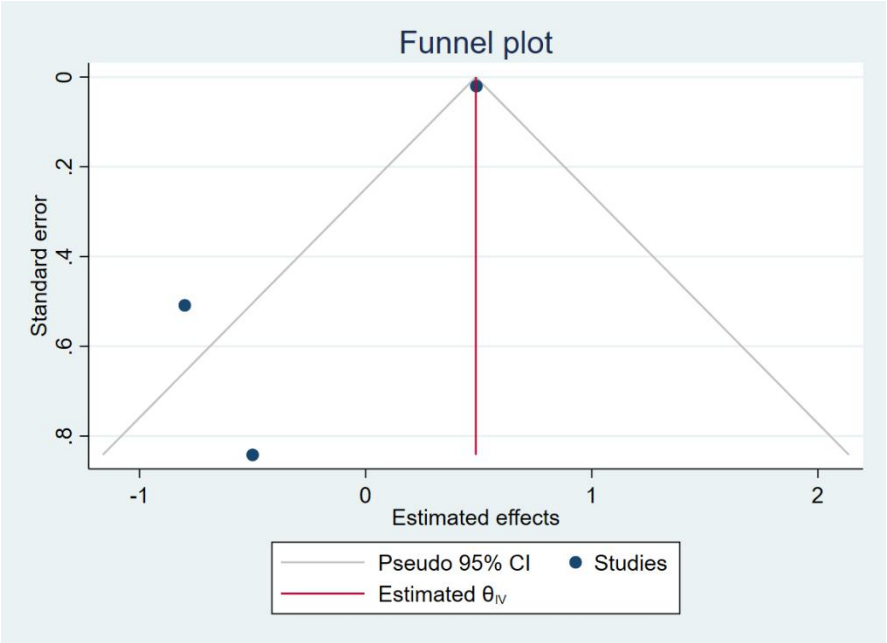


Figure S6. Funnel plot analysis in the meta-analysis of the association between short-term exposure to PM_{10} and DBP.

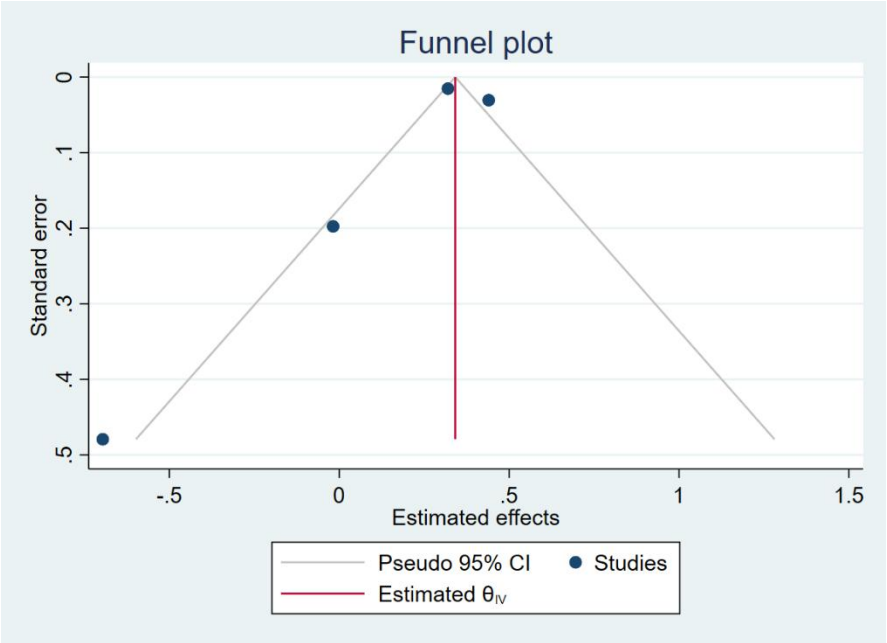


Figure S7. Forest plot for the association between long-term exposure to $PM_{2.5}$ (per 10 μg/m^3 increment) and SBP (mmHg).

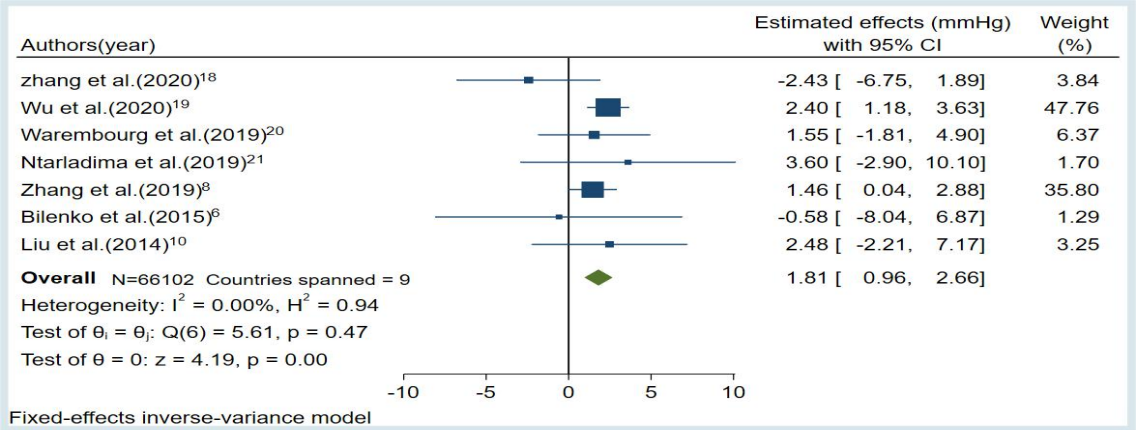


Figure S8. Forest plot for the association between long-term exposure to PM_{10} (per 10 μg/m^3 increment) and SBP (mmHg).

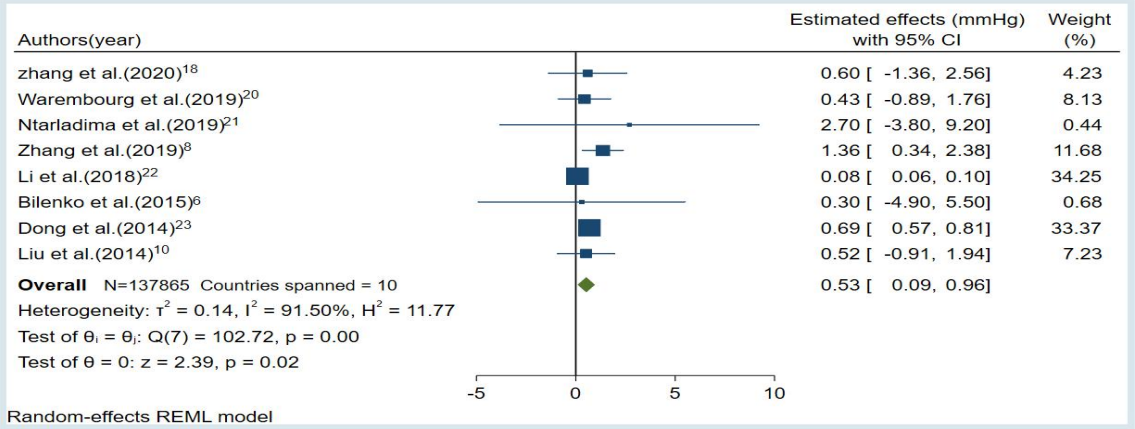


Figure S9. Forest plot for the association between long-term exposure to NO_2 (per 10 μg/m^3 increment) and SBP (mmHg).

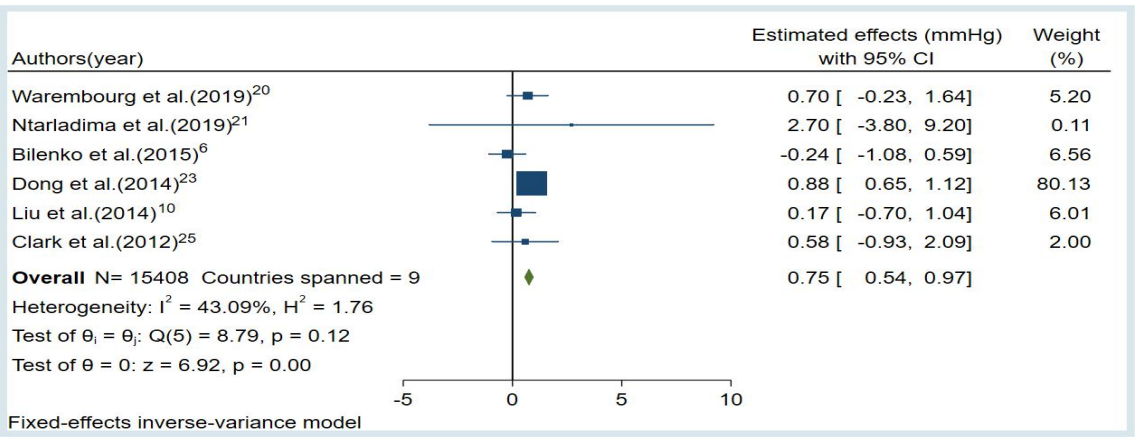


Figure S10. Forest plot for the association between long-term exposure to $PM_{2.5}$ (per 10 μg/m^3 increment) and DBP (mmHg).

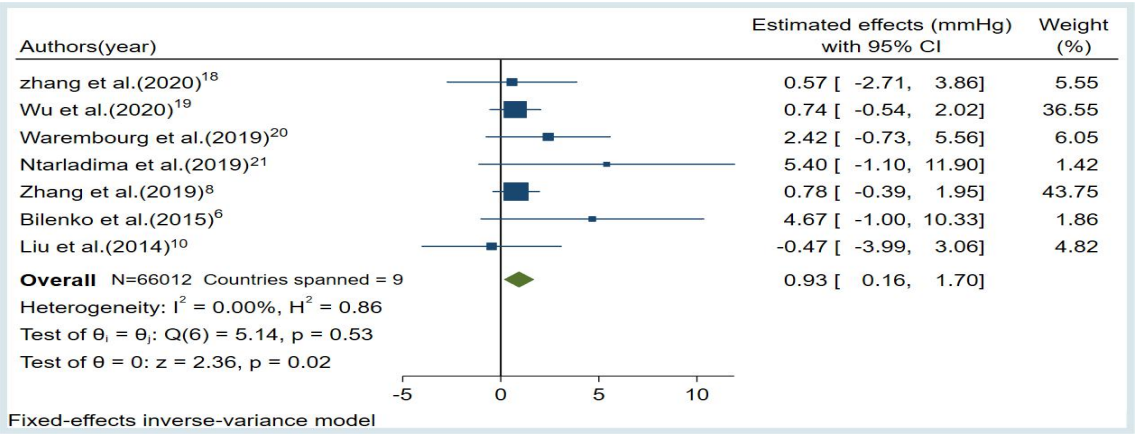


Figure S11. Forest plot for the association between long-term exposure to PM_{10} (per 10 $\mu g/m^3$ increment) and DBP (mmHg).

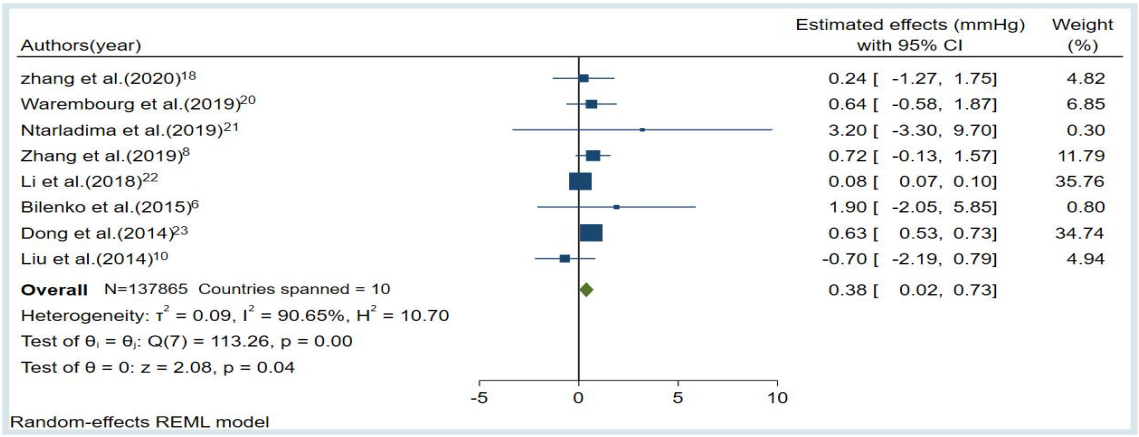


Figure S12. Forest plot for the association between long-term exposure to NO_2 (per 10 $\mu g/m^3$ increment) and DBP (mmHg).

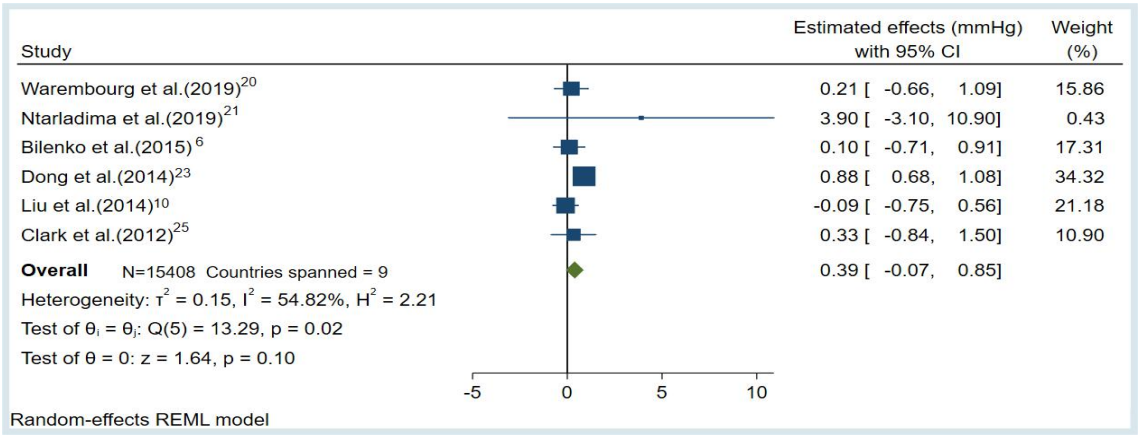


Figure S13. Funnel plot analysis in the meta-analysis of the association between long-term exposure to $PM_{2.5}$ and SBP.

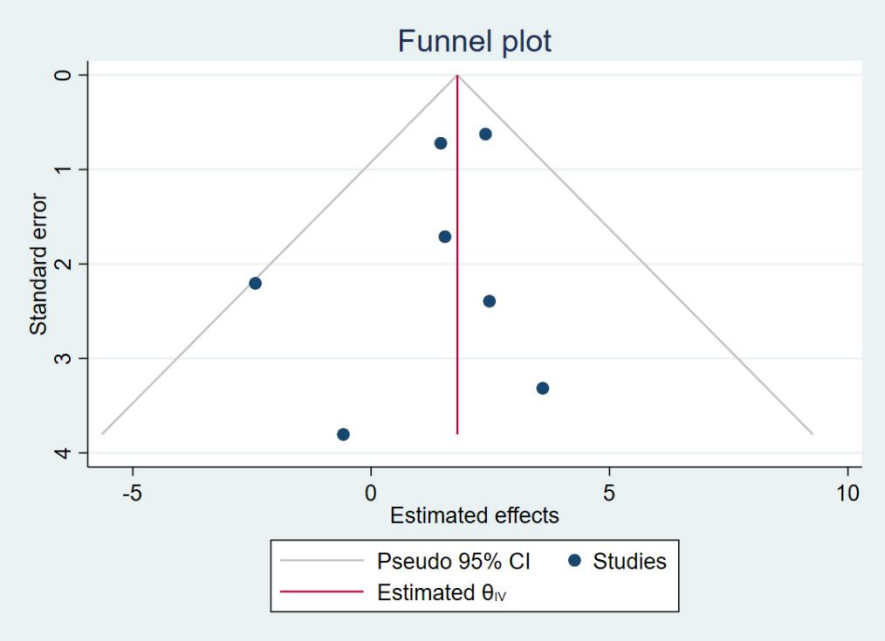


Figure S14. Funnel plot analysis in the meta-analysis of the association between long-term exposure to $PM_{2.5}$ and DBP.

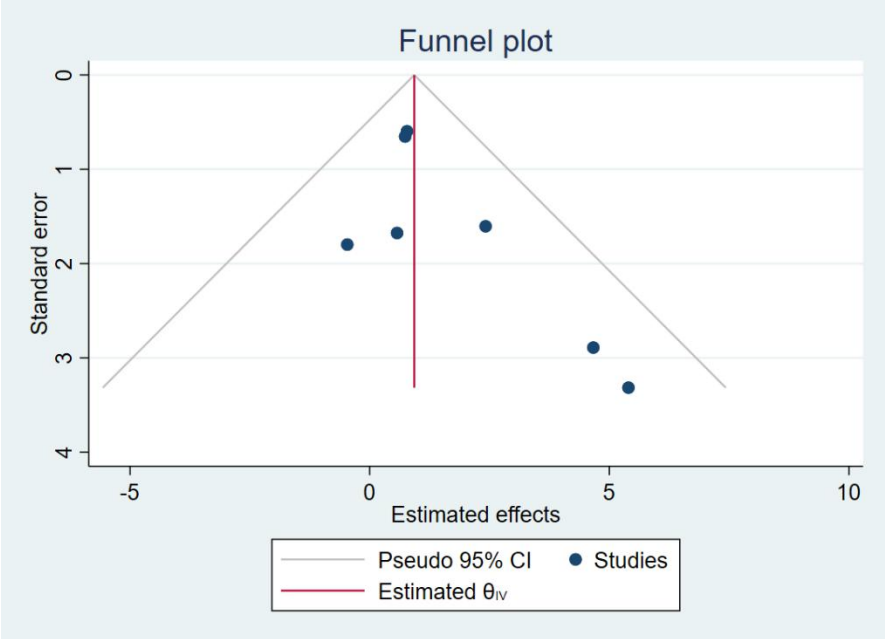


Figure S15. Funnel plot analysis in the meta-analysis of the association between long-term exposure to PM_{10} and SBP.

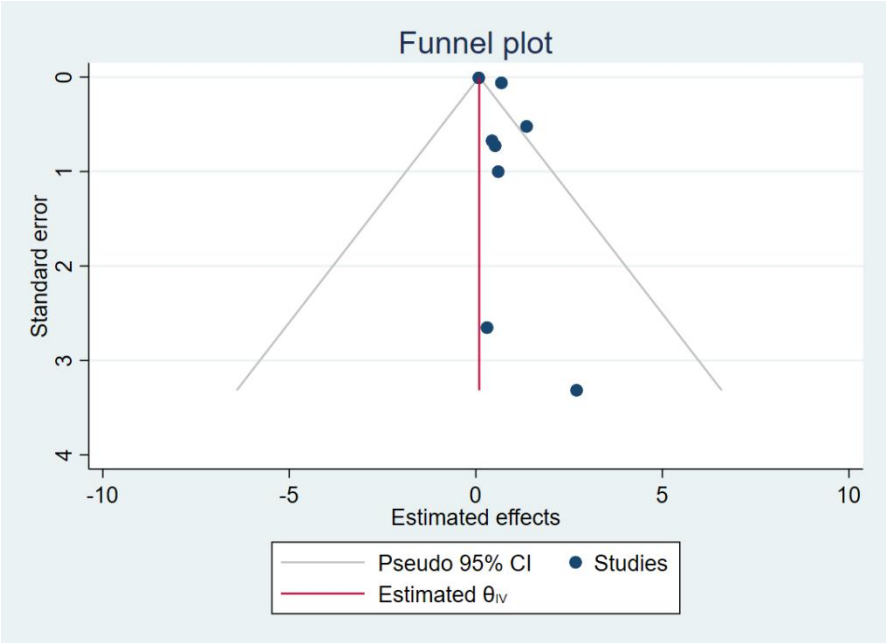


Figure S16. Funnel plot analysis in the meta-analysis of the association between long-term exposure to PM_{10} and DBP.

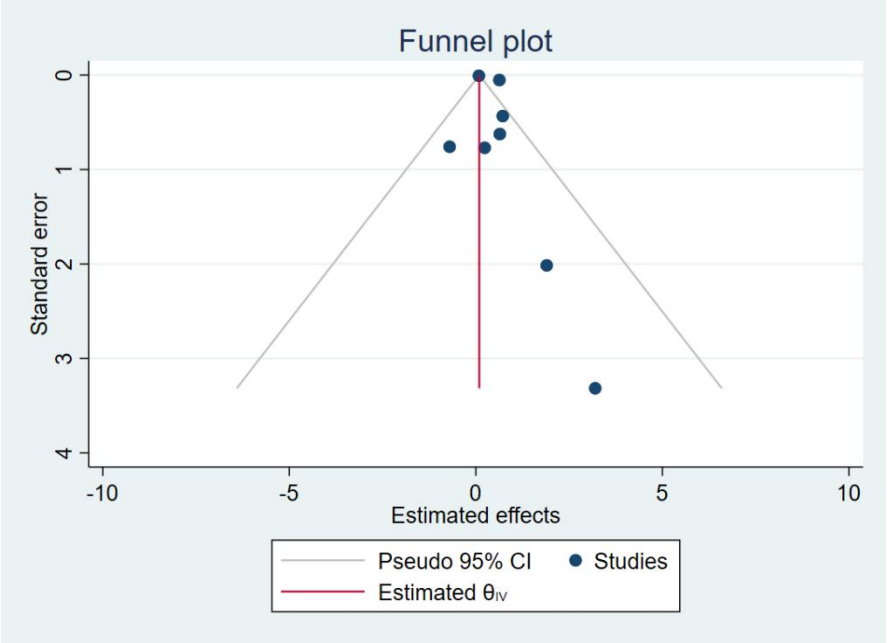


Figure S17. Funnel plot analysis in the meta-analysis of the association between long-term exposure to NO_2 and SBP.

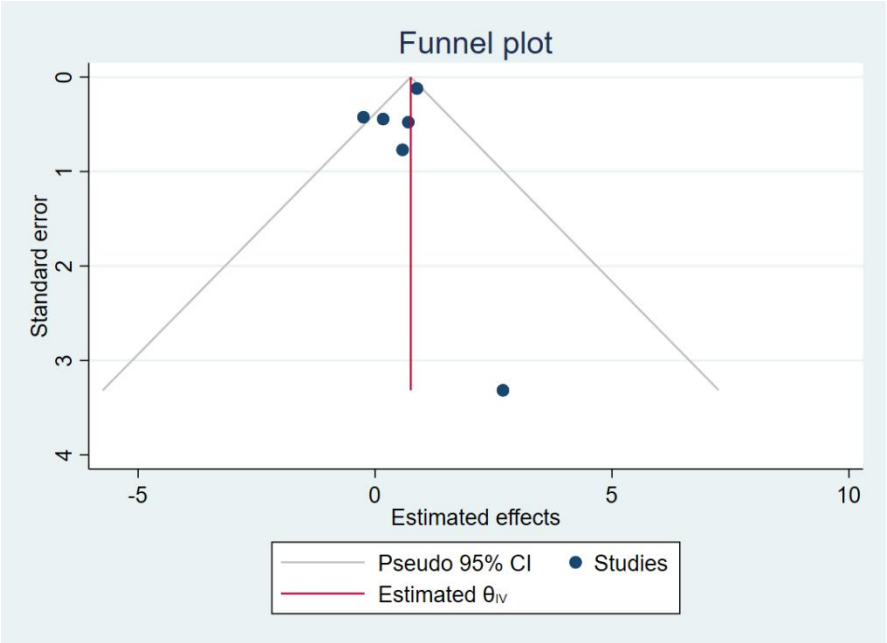


Figure S18. Funnel plot analysis in the meta-analysis of the association between long-term exposure to NO_2 and DBP.

